

Binance Crypto Alpha Research

1. Introduction

- **Systematic mid-frequency statistical-arbitrage** on the Binance perpetual-futures market.
- Trades a broad cross-section of liquid USDT-margined perpetual contracts on a **daily rebalancing cycle**.
- Long and short positions sized to be **market-neutral** and dollar-balanced at portfolio level.
- Each contract enters the tradable universe only after passing a **quarterly liquidity and continuity filter**; the universe rolls forward through the backtest so that delisted or freshly-listed contracts are added and removed at the right point in time, preventing survivorship bias.

Market	Binance USDT-margined perpetual futures
Frequency	Mid-frequency, daily rebalance
Style	Cross-sectional statistical arbitrage, market-neutral
Universe	150 symbols, rolling quarterly refresh
Backtest window	20240101 — 20260101
Cost model	5 bps fee + 2 bps slippage
Data sources	Binance Futures REST (price, volume), funding-rate history, open-interest archive, top-trader long/short ratio archive

2. Signal Pipeline

Three independent signals feed a portfolio combiner. All three share two construction invariants that are applied at every stage of the pipeline:

- **Cross-sectional neutralisation.** Every raw signal is residualised against a rolling market beta before any portfolio construction. The model therefore trades idiosyncratic dispersion rather than leveraged market exposure.
- **Persistent long/short balance.** The weight construction enforces approximately equal gross capital on the long and short sides each day, so the strategy retains its market-neutral profile across regime shifts rather than drifting net long in bull markets or net short in drawdowns.

2.1 Momentum

- Cross-sectional preference for assets whose price trend is supported by an **expansion in open interest** — positions accumulating in the direction of the move.
- **Funding-rate filter** down-weights signals when the funding side disagrees with the price direction (a crowded-trade flag).
- **Volatility-rank adjustment** shrinks positions in unusually noisy names.

2.2 Liquidation Reversal

- Targets dislocations caused by **liquidation cascades**: an unusually large one-day drop in open interest is a forced-position flush that tends to mean-revert within a few days.
- **Liquidation shock** component flags the abnormal OI move.
- **Regime score** encompasses long/short ratio change to indicate shifts in market regime.
- Event-driven and naturally **counter-cyclical to the momentum strategy**.

2.3 EBM Machine-Learning Signal

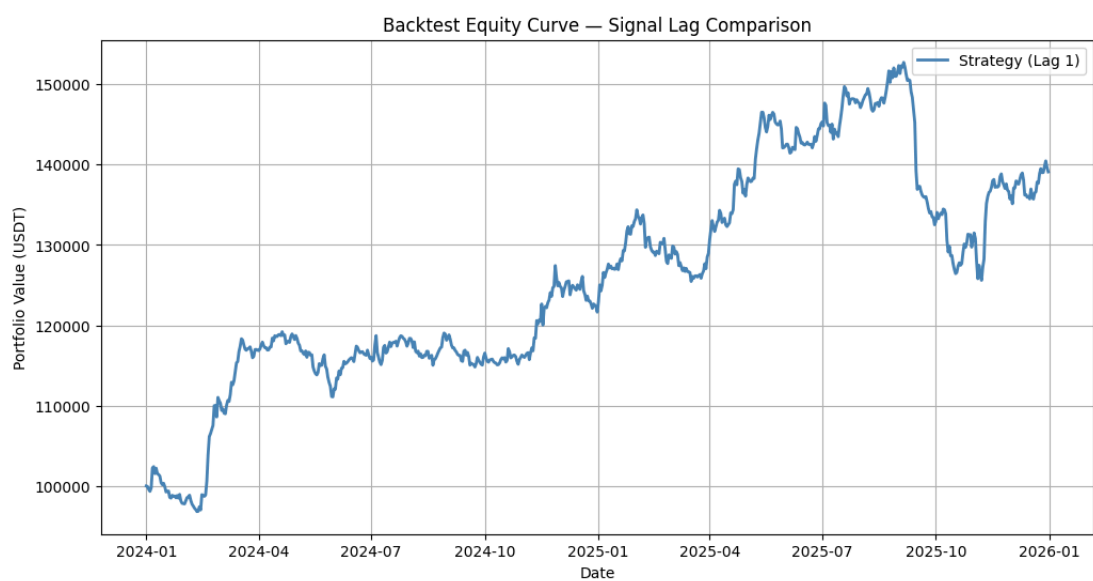
- **Explainable Boosting Machine** trained walk-forward on the full factor panel.
- Non-parametric in its feature relationships — discovers piecewise shape functions and pairwise interactions rather than imposing them.
- Global model is trained with factors **neutralised with respect to the regime separator**, which is market volatility in our experiment.
- A **mixture-of-experts** extension trains a per-regime expert on the global model's residuals so that the residual structure inside each regime can be captured.
- The model is trained only with factors **passing shape-health analysis**.

2.4 Portfolio Combiner

- Daily **mean-variance optimisation** in strategy-return space.
- **EMA smoothing** as a post-process on the final signal.
- **Ledoit-Wolf shrinkage** stabilises the covariance matrix, which is otherwise unstable because the number of signals is small.

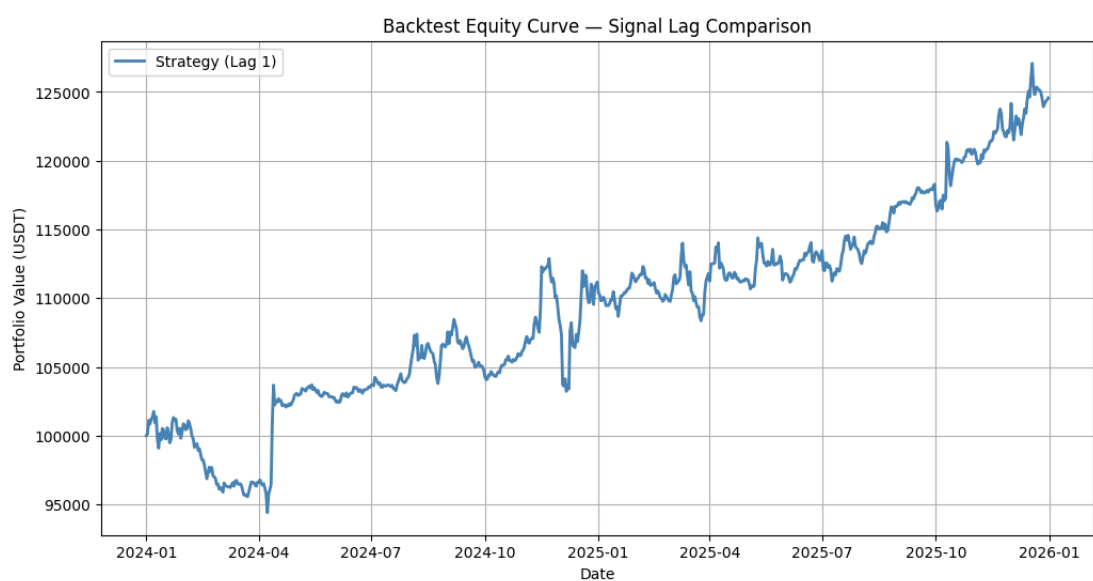
3. Performance

3.1 Liquidation Reversal



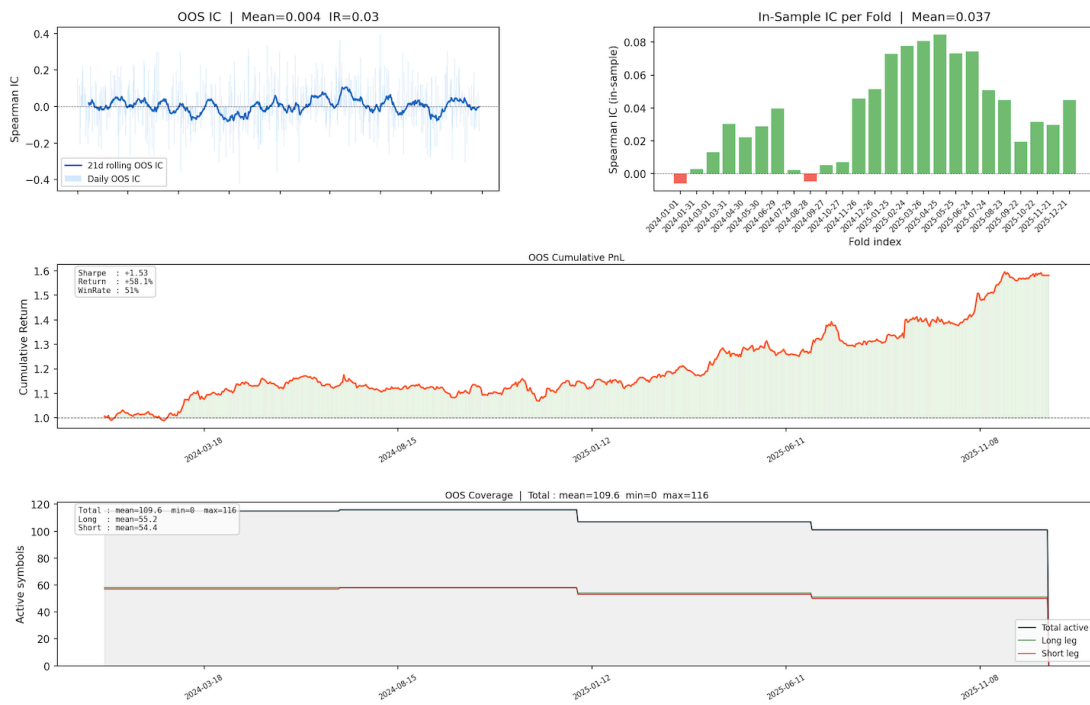
Final return	39.1 %
Annualised Sharpe	1.35
PSR (SR* = 0)	0.9923

3.2 Momentum



Final return	24.5 %
Annualised Sharpe	1.02
PSR (SR* = 0)	98.2

3.3 EBM ML Signal

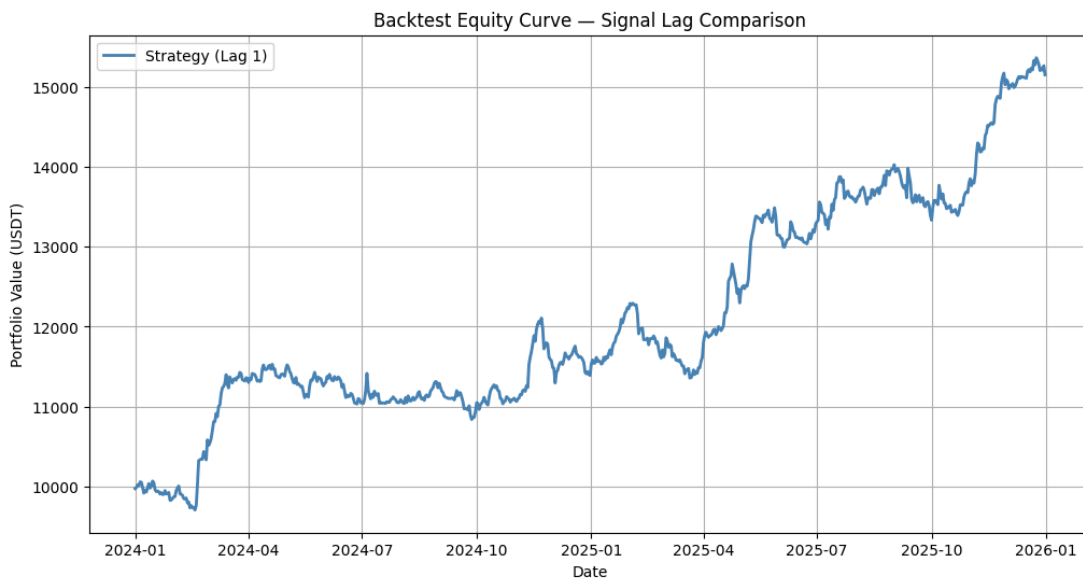


- **Top:** per-fold feature importance.
- **Bottom-left:** in-sample vs out-of-sample information coefficient over time.
- **Bottom-right:** cumulative PnL of the EBM signal in isolation.

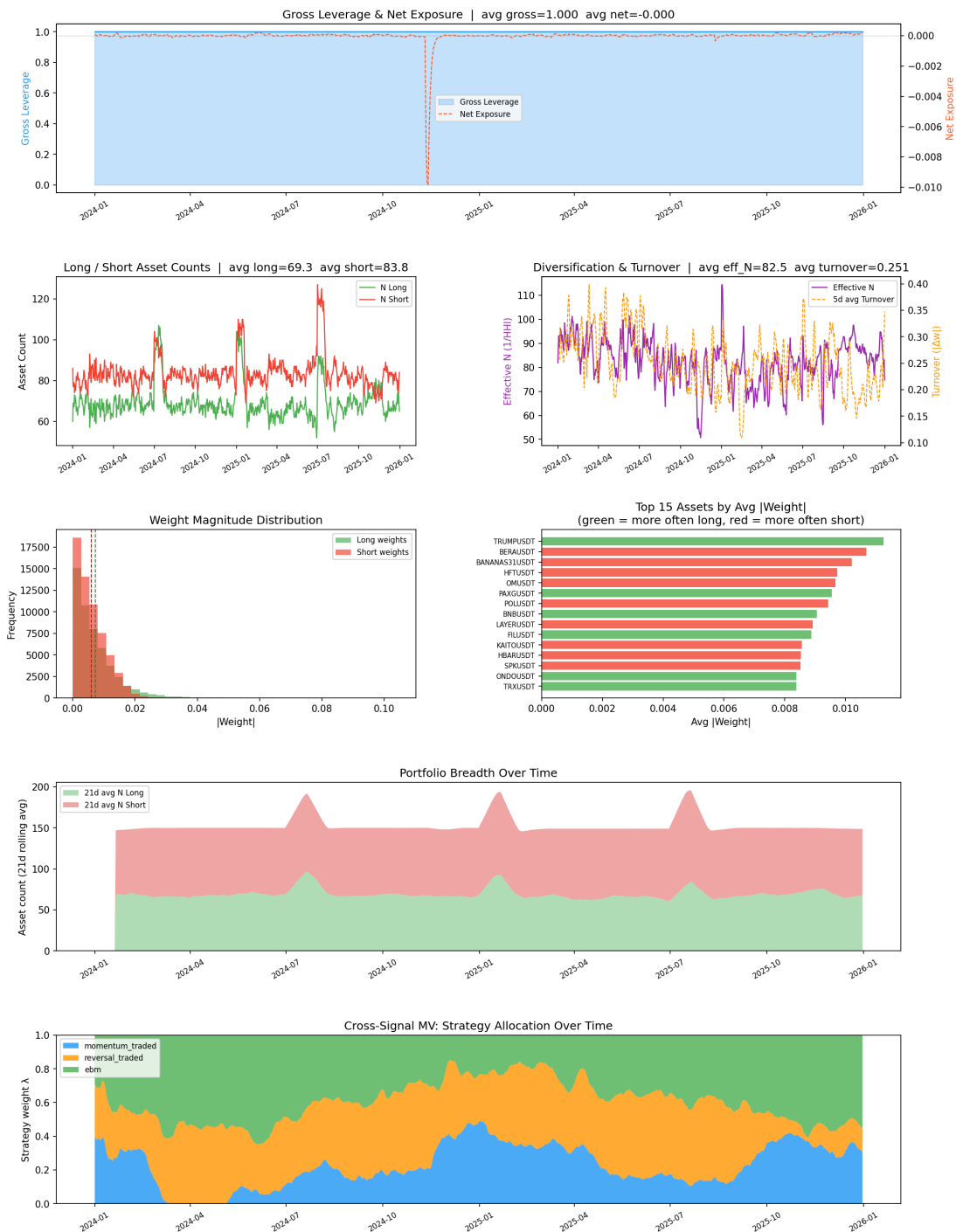
Final return	TODO
Annualised Sharpe	TODO
PSR (net)	TODO
Average daily turnover	TODO

3.4 Combined Portfolio

- Blends the three signals through the strategy-space mean-variance optimiser.



Weight Distribution Analysis | method=cross_signal_mv



- The lower panel reports **gross leverage** and **net exposure**, **long-short asset counts**, the **effective number of positions**, **daily turnover**, the **weight-magnitude histogram**, and the **most-traded symbols**.
- Net exposure tracks zero and long-short asset counts move in lock-step — the persistent market-neutral and dollar-balanced behaviour described above.

Total return	51.2 %
Annualised Sharpe	1.91
PSR (SR* = 0)	.9974